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Embedded System Architecture - CSEN 701

Module 2: *Microcontroller Fundamentals*

Lecture 04: *Overview on ARM Cortex M0+ Architecture*

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Outline

- Recap on RP2040
- The ARM Processors
- ARM Cortex M0+

Recap on RP2040

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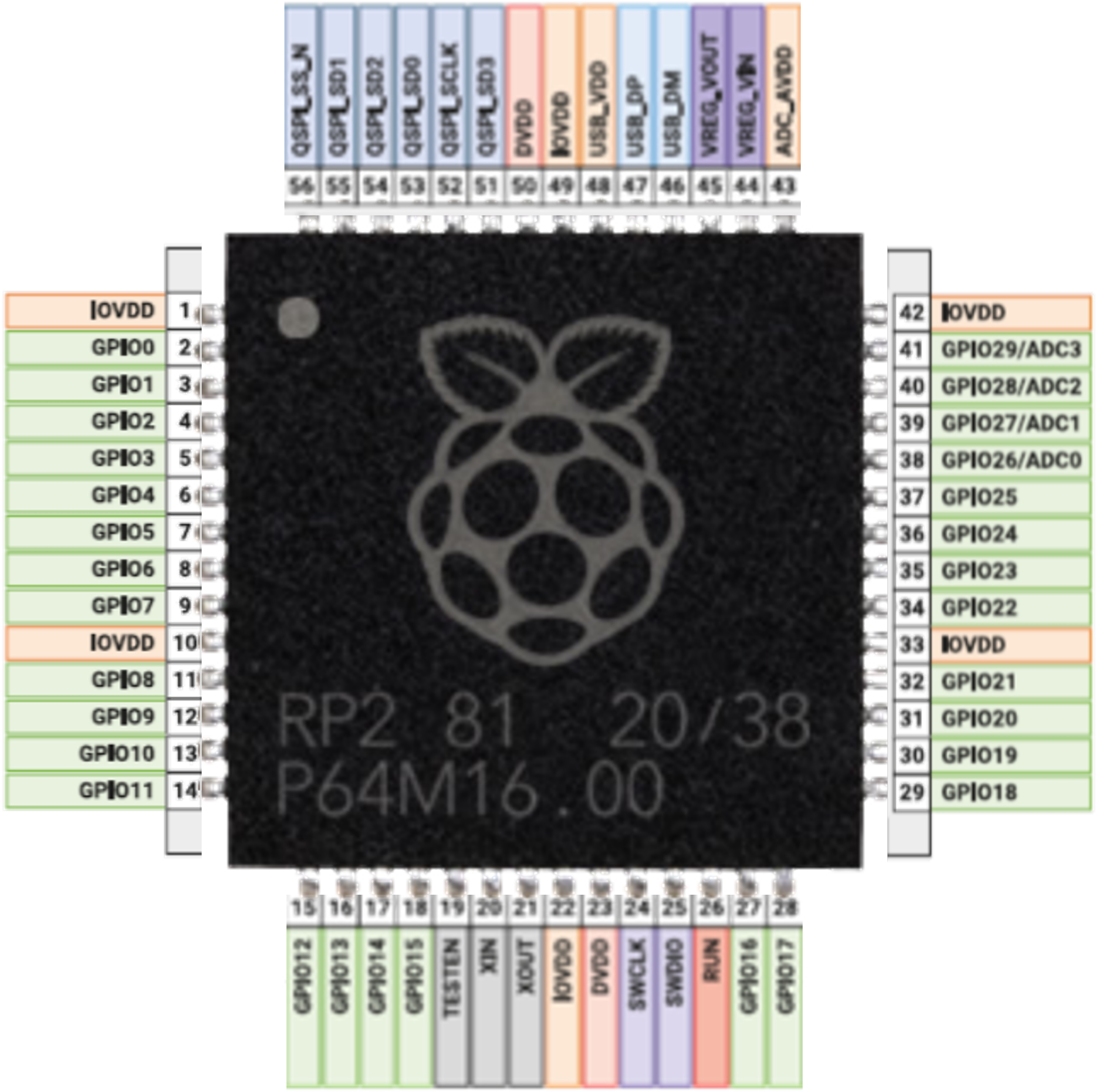
System Overview of the RP2040 Chip



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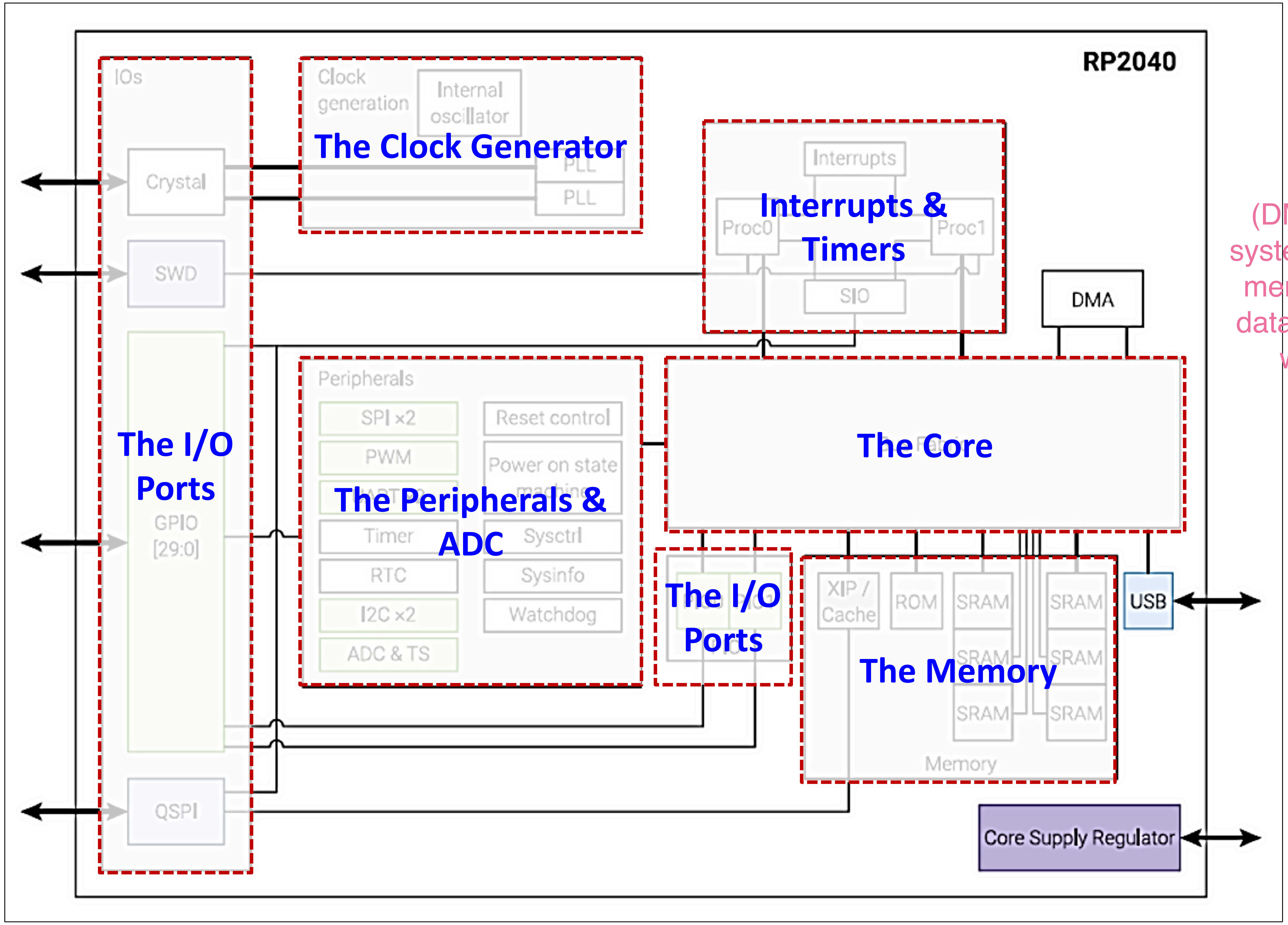
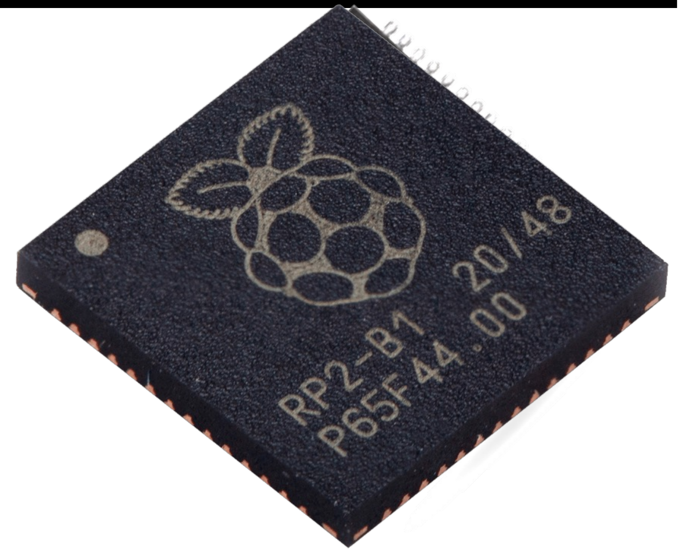
System Overview of the RP2040 Chip

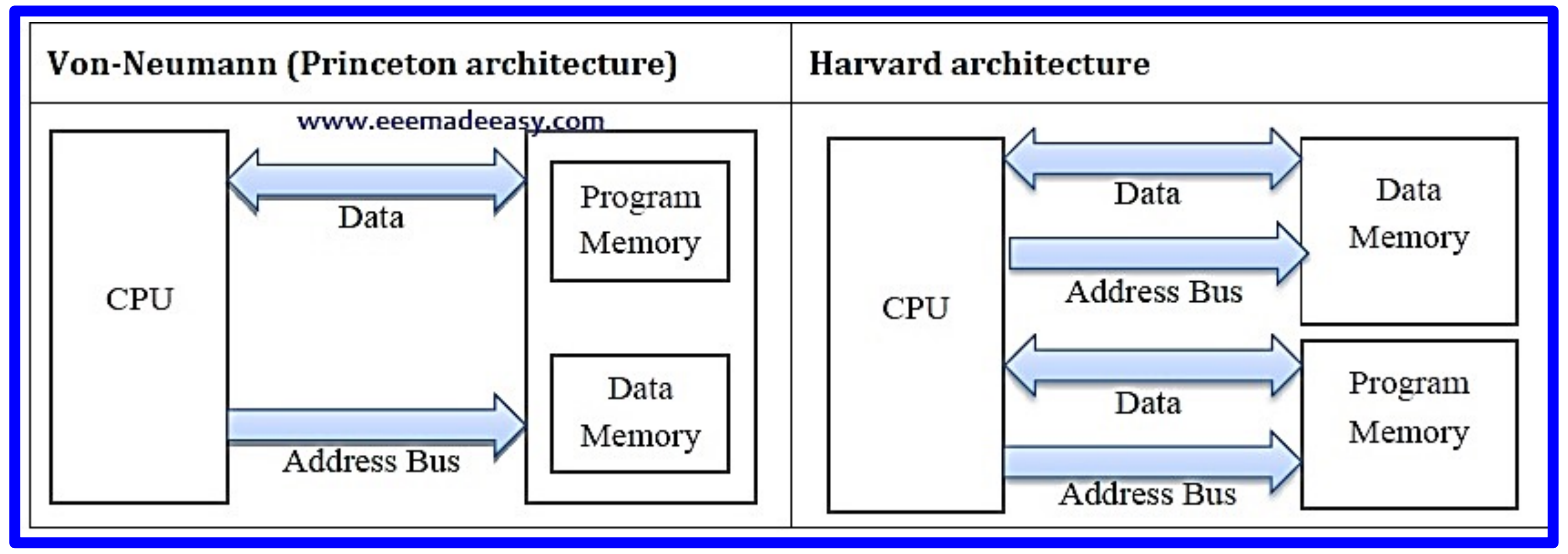


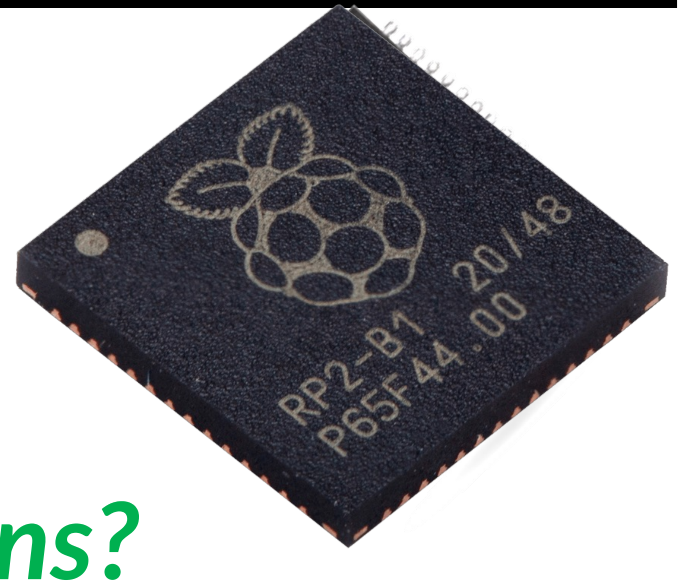


(DMA) is a feature in computer systems that allows peripherals or memory components to transfer data directly to and from memory without involving the CPU.

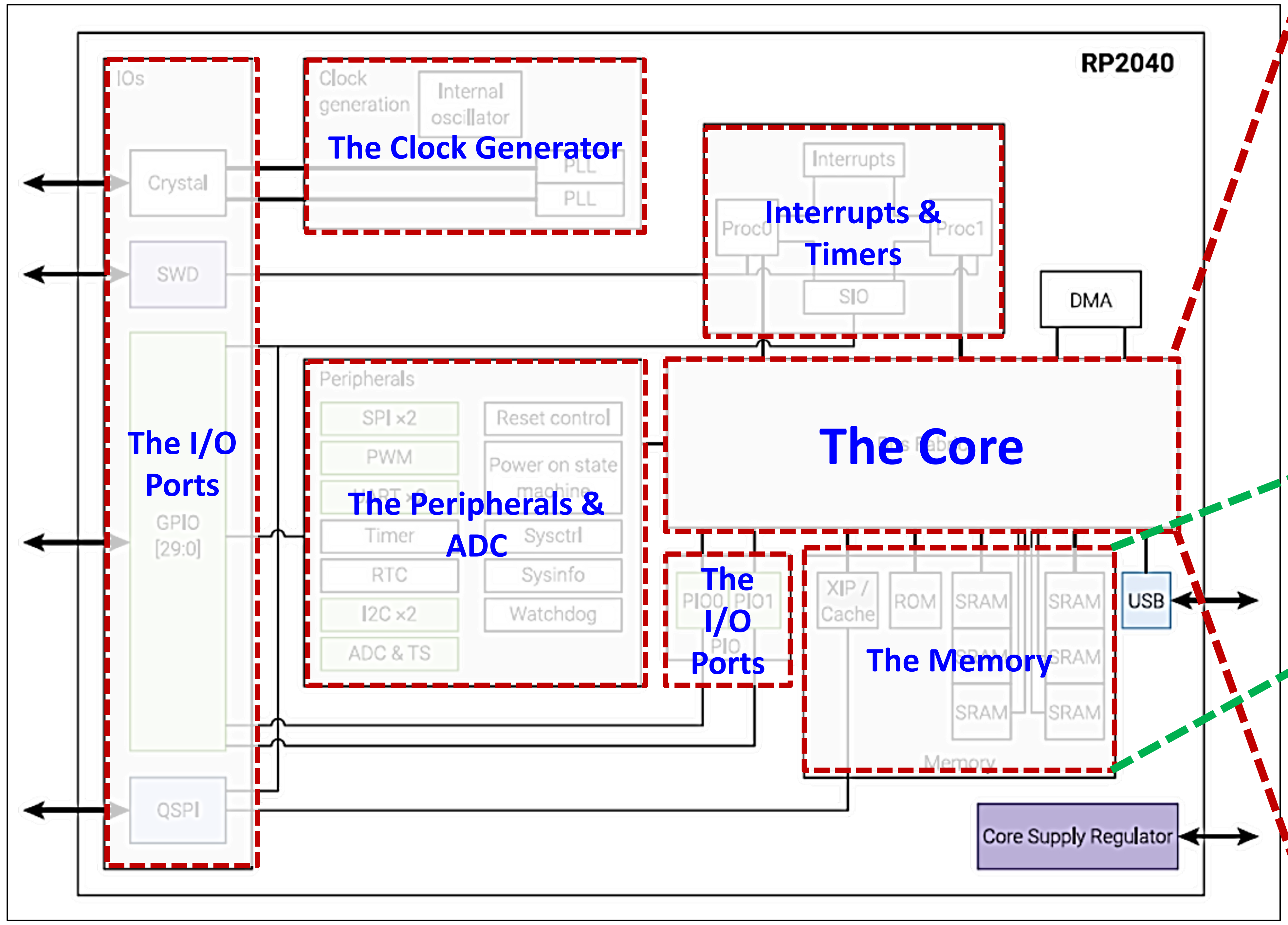
Recap on RP2040

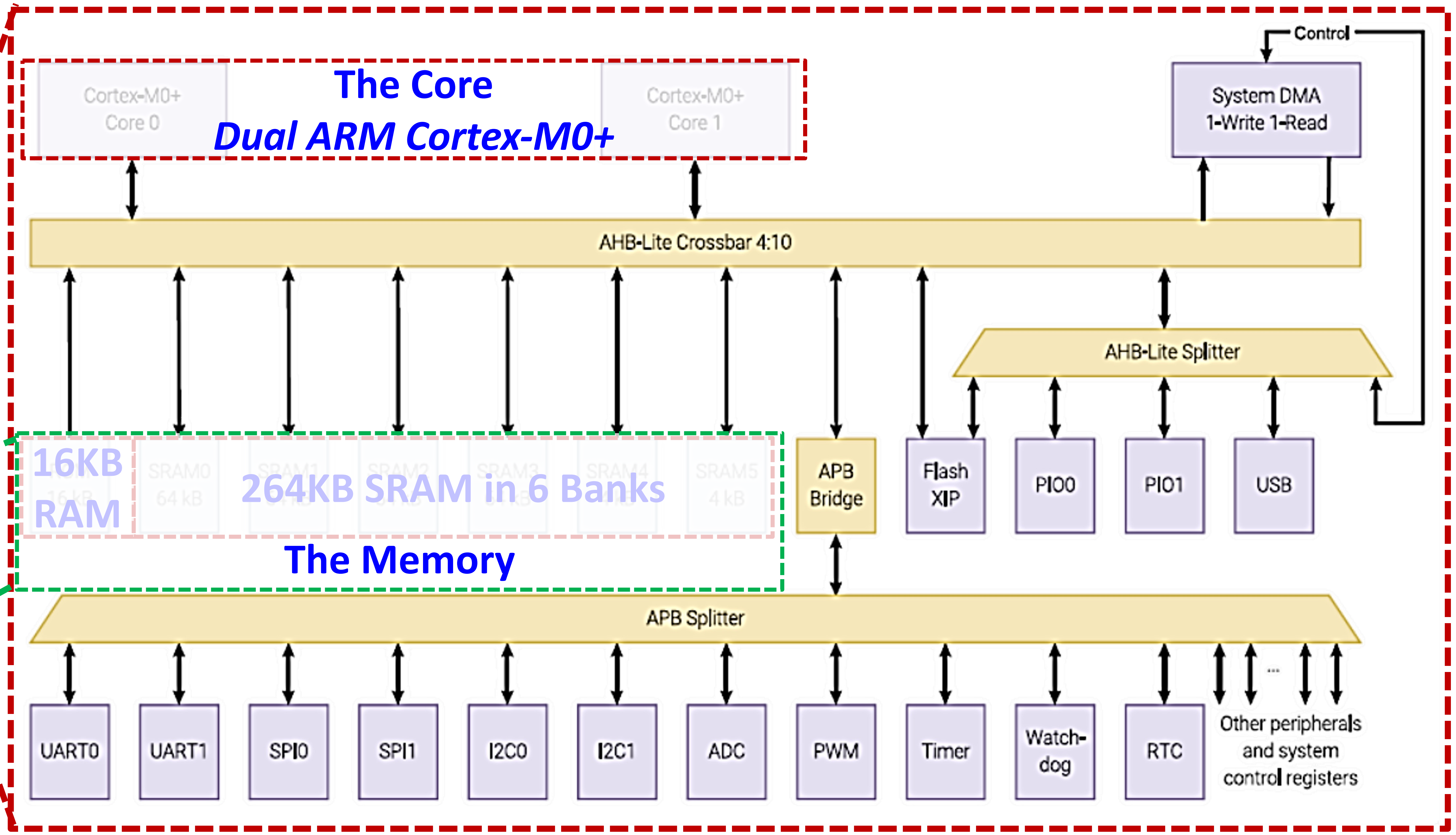
System Overview of the RP2040 Chip





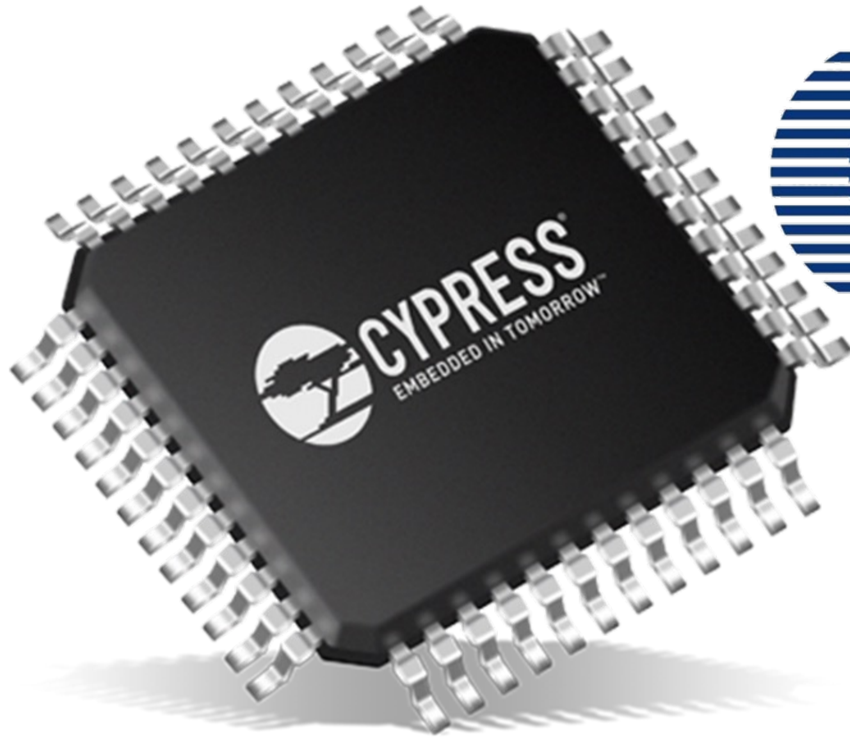
Any Observations?

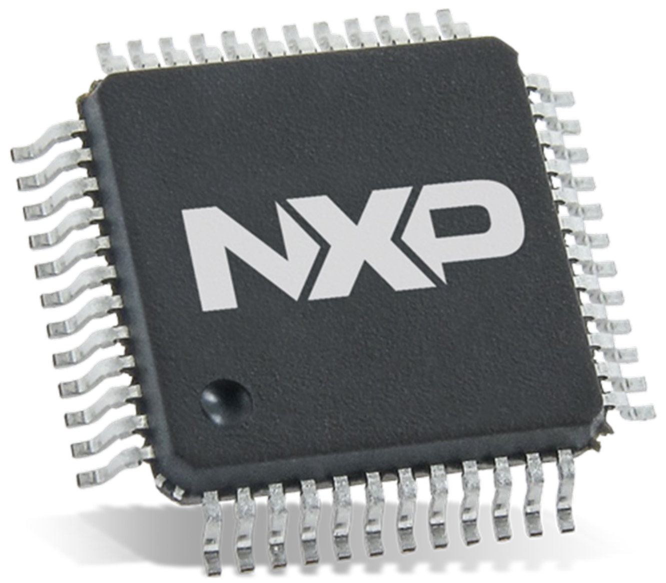


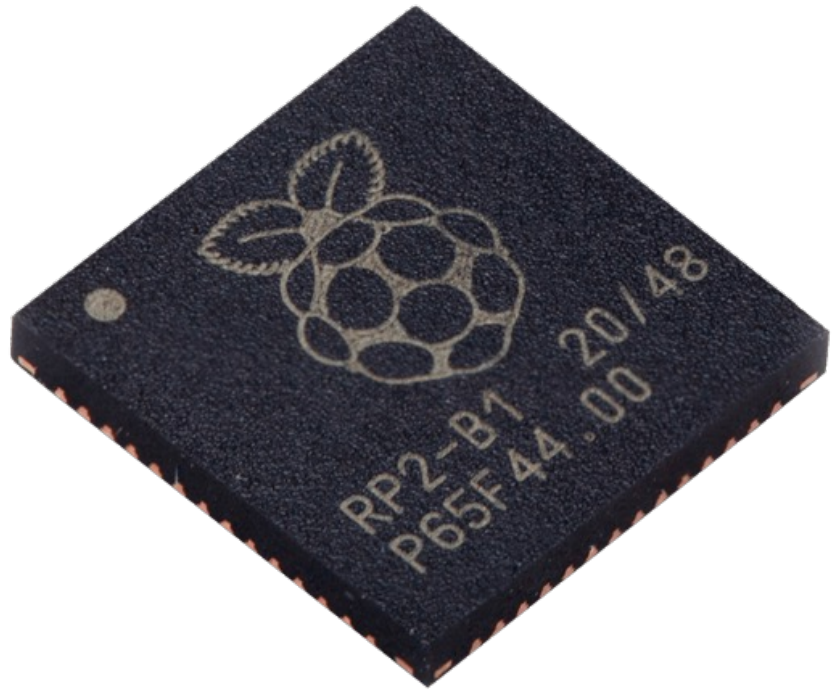


The ARM Processors

Arm Microcontrollers Manufacturers







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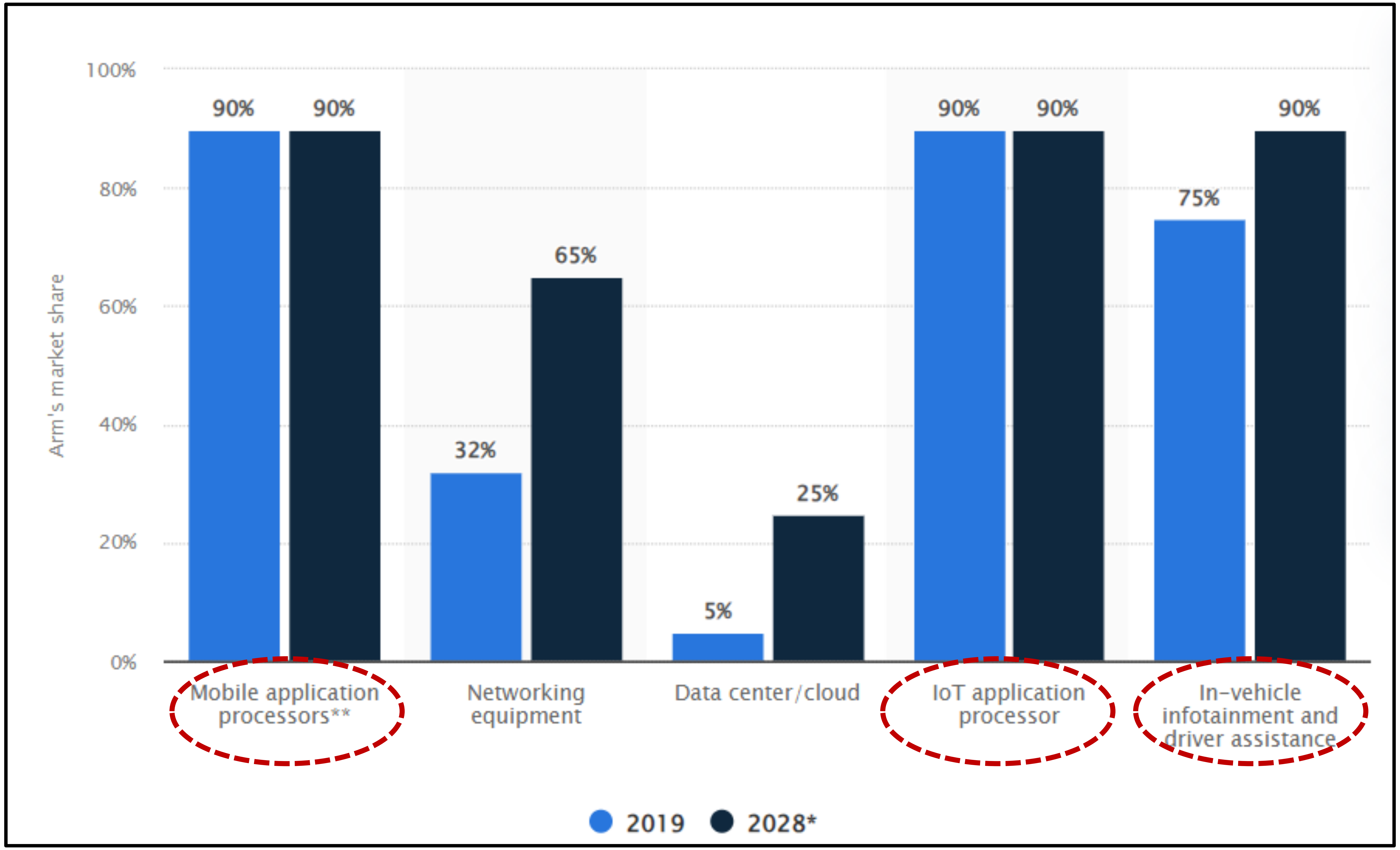
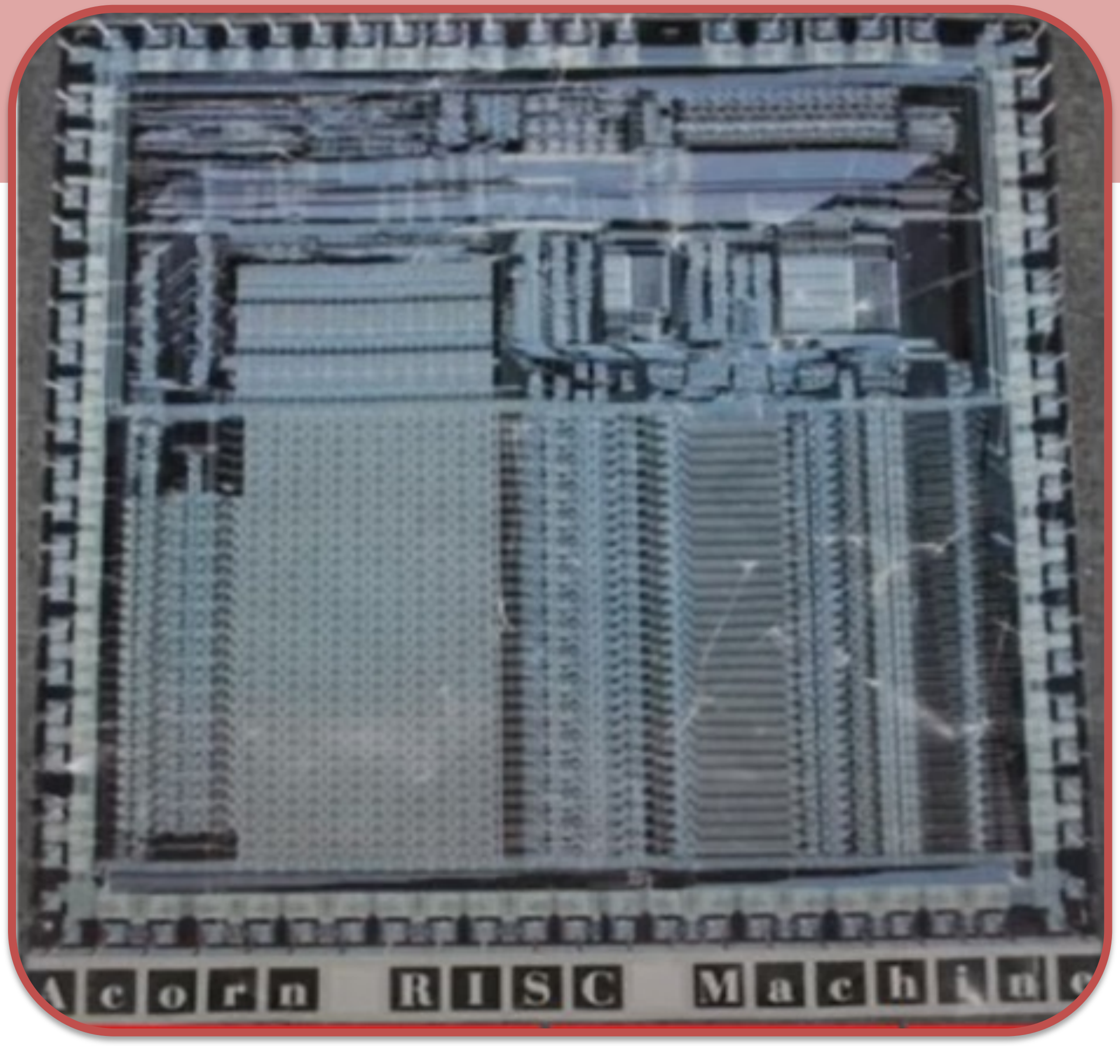
Arm Market Share!!



Some History!

1983-1985

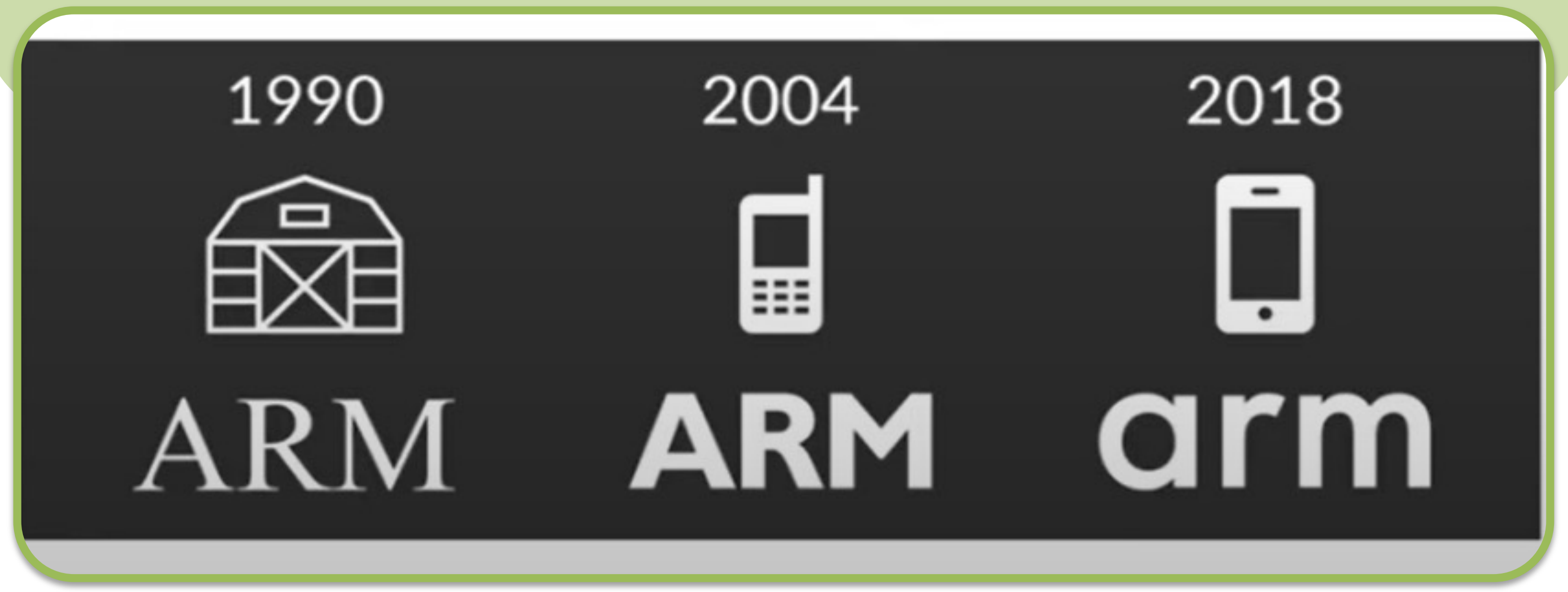
- The first Arm is developed at Acorn Computers limited at Cambridge, England
- **ARM** Stands for **_A_**corn **_R_**ISC **_M_**achine



1990

- Company has structured as joint between Acorn Computers, Apple Computers (Apple Inc.) and VLSI Technology.
- The name has changed to

_A_dvanced **_R_**ISC **_M_**achine



Why ARM?

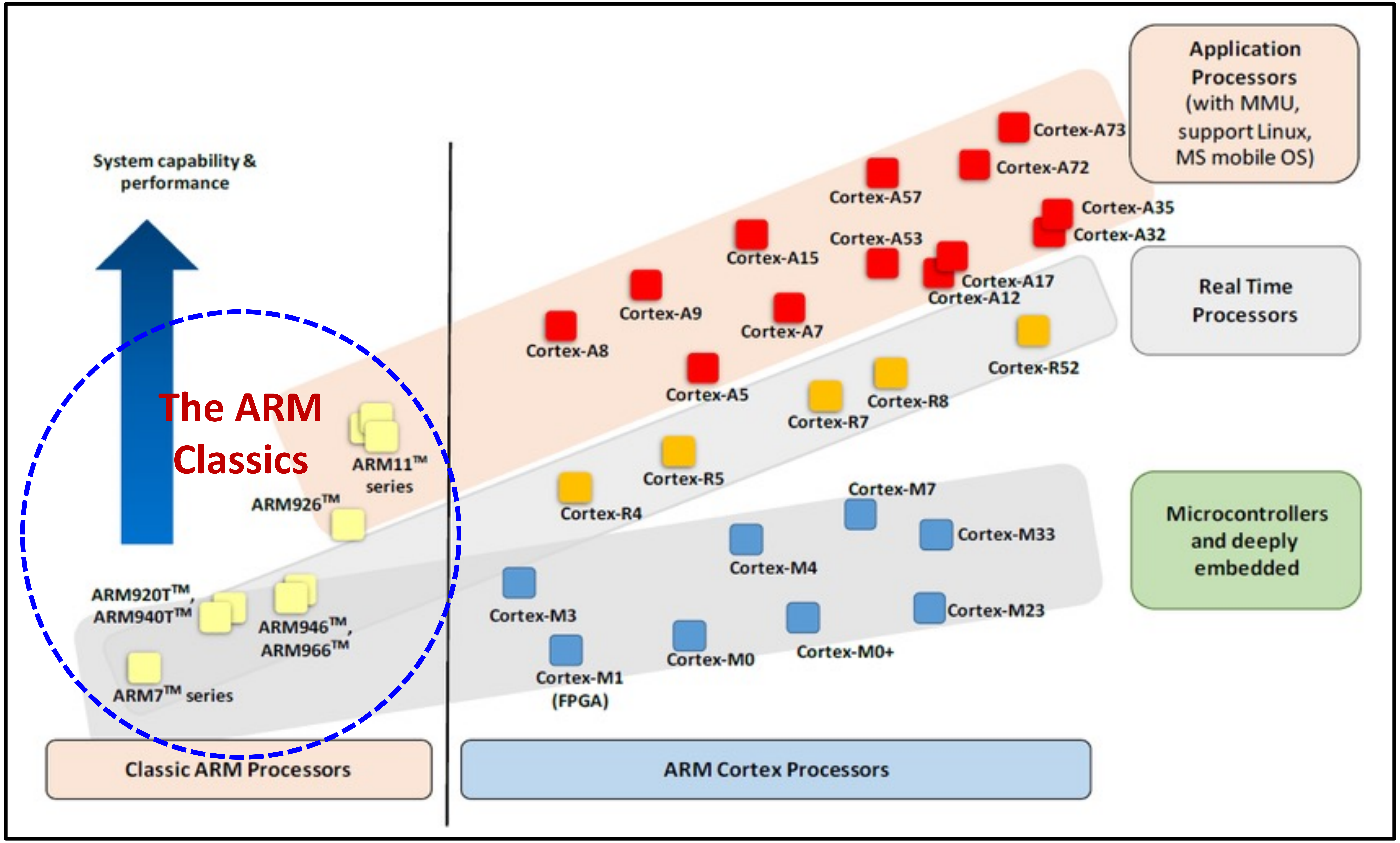
- ARM Architecture is a family of RISC-based Processor architectures:
  - ➢ Power Efficiency, because it requires less transistors per instruction in RISC
  - ➢ Widely used in mobile devices,
  - ➢ Designed and licensed to a wide eco-system by ARM

ARM Processors: Classes

A R M

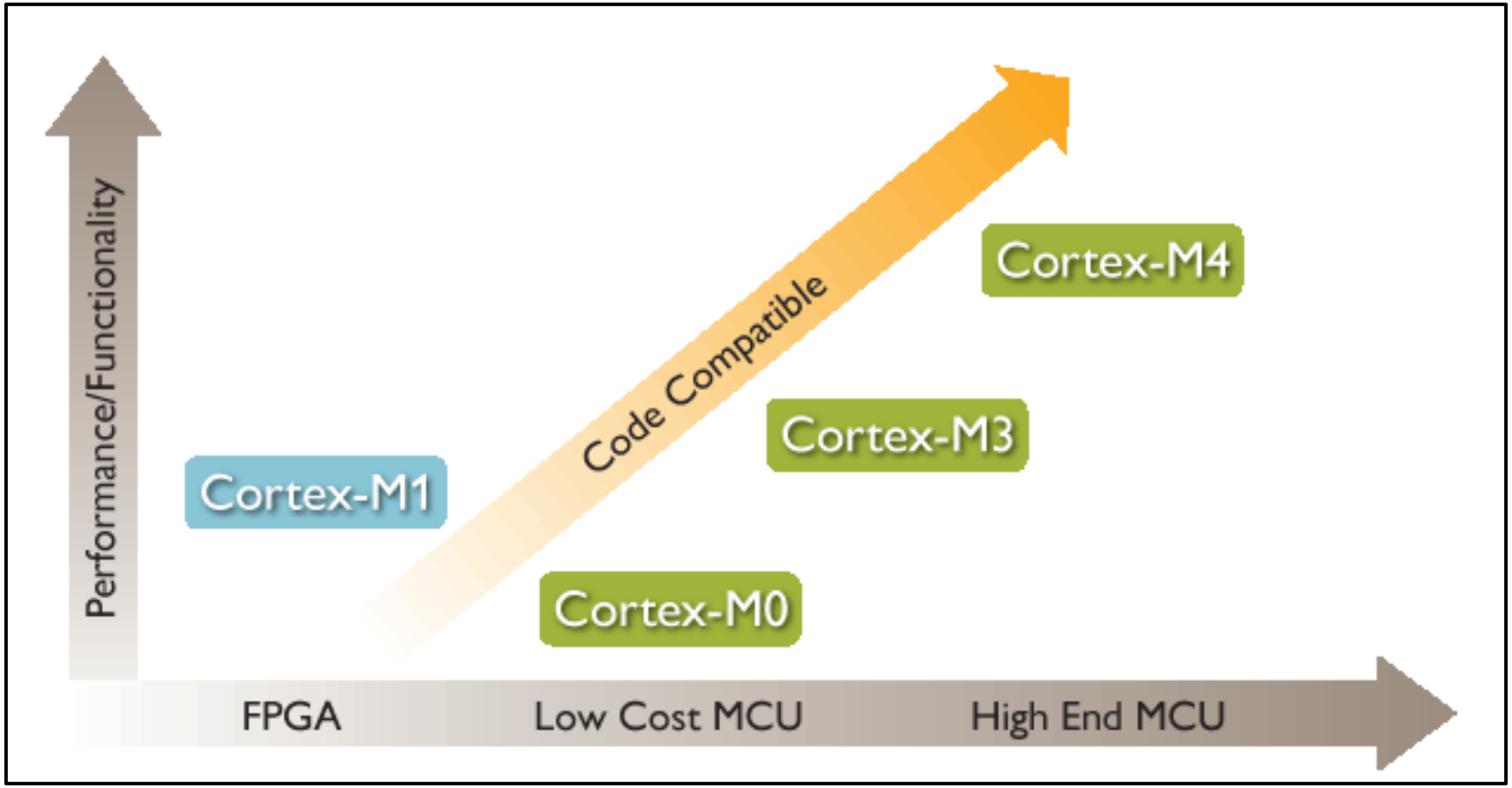
CORTEX-Ax	CORTEX-Rx	CORTEX-Mx
Application	**Real-time**	**Microcontroller**

ARM Processors: Classes Performance

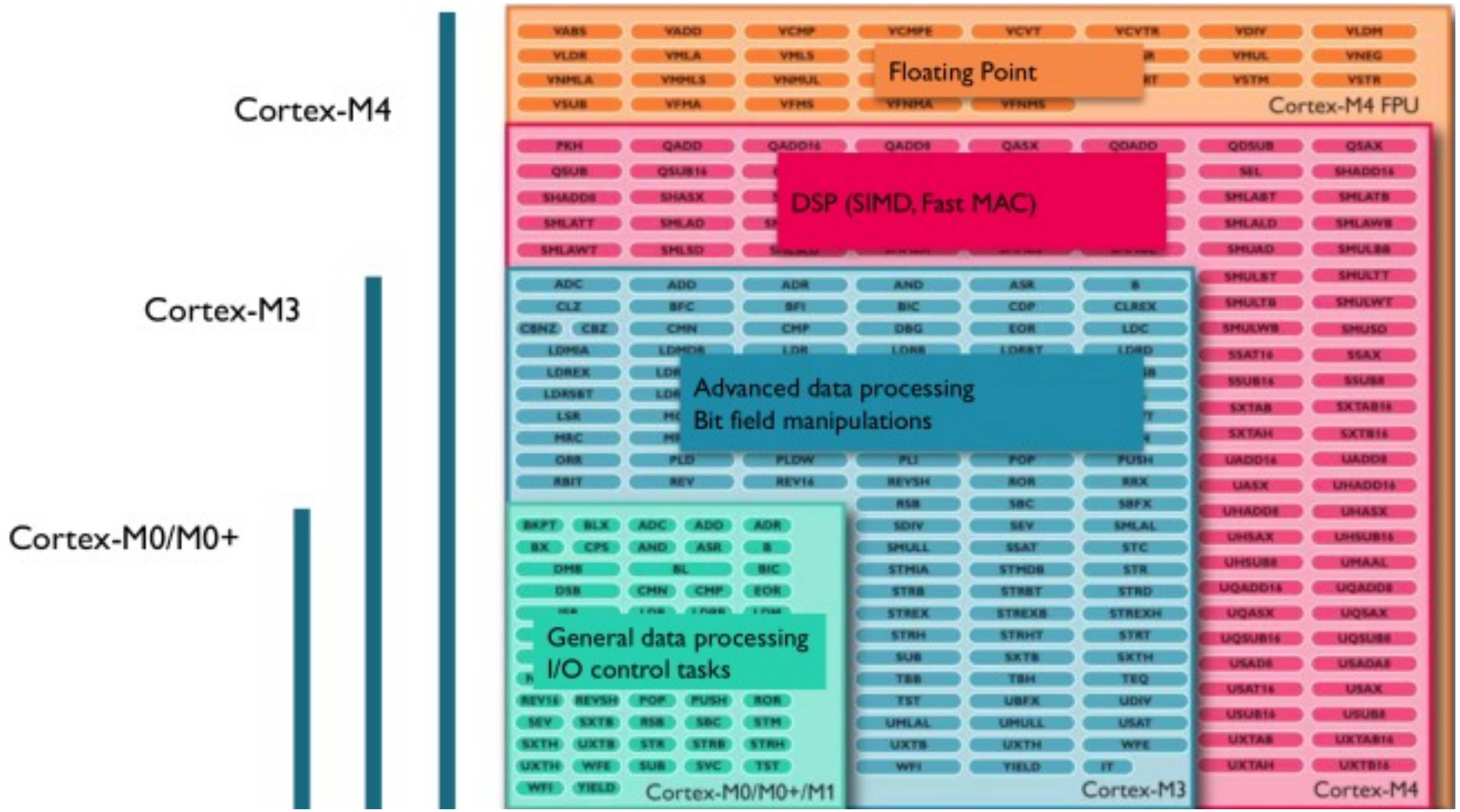


ARM Processors: Classes Applications

AMR Cortex-Mx: Classes Performance



AMR Cortex-Mx: Instruction Set Architecture



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AMR Cortex-Mx: Instruction Set Architecture

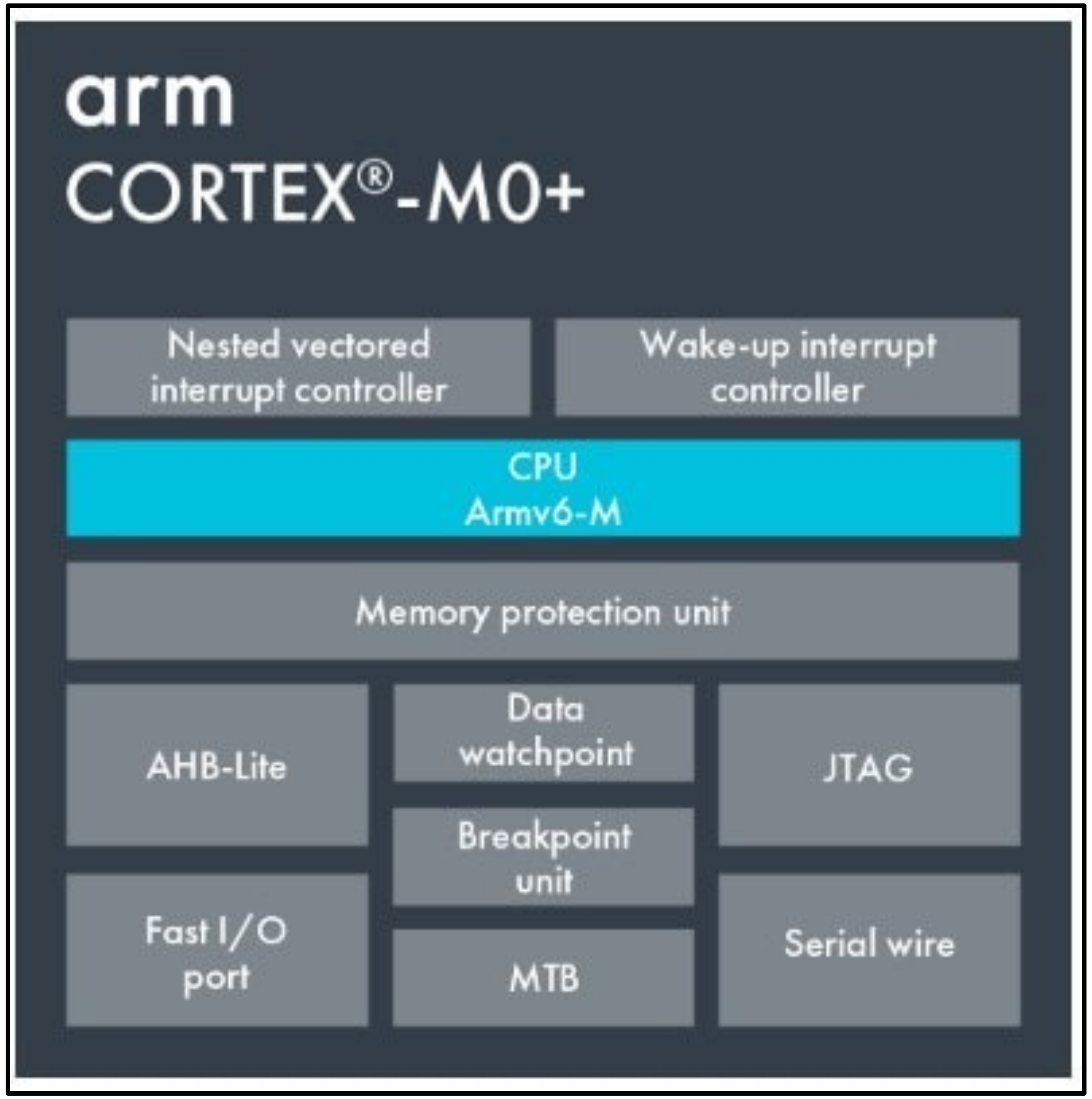
Processor	ARM Architecture	Core Architecture	Thumb®	Thumb®-2	Hardware Multiply	Hardware Divide	Saturated Math	DSP Extensions	Floating Point
Cortex-M0	ARMv6-M	Von Neumann	Most	Subset	Optional	No	No	No	No
Cortex-M0+	ARMv6-M	Von Neumann	Most	Subset	Optional	No	No	No	No
Cortex-M1	ARMv6-M	Von Neumann	Most	Subset	Optional	No	No	No	No
Cortex-M3	ARMv7-M	Harvard	Entire	Entire	Optional	Yes	Yes	No	No
Cortex-M4	ARMv7E-M	Harvard	Entire	Entire	Optional	Yes	Yes	Yes	Optional

Overview

- The Cortex-M0+ processor builds on the very successful Cortex-M0 processor, retaining full instruction set and tool compatibility, while further **reducing energy consumption** and increasing performance.

- Remember that the Cortex-M0+ itself is a ***microprocessor***.

- The main architecture is the **ARMv6-M**.



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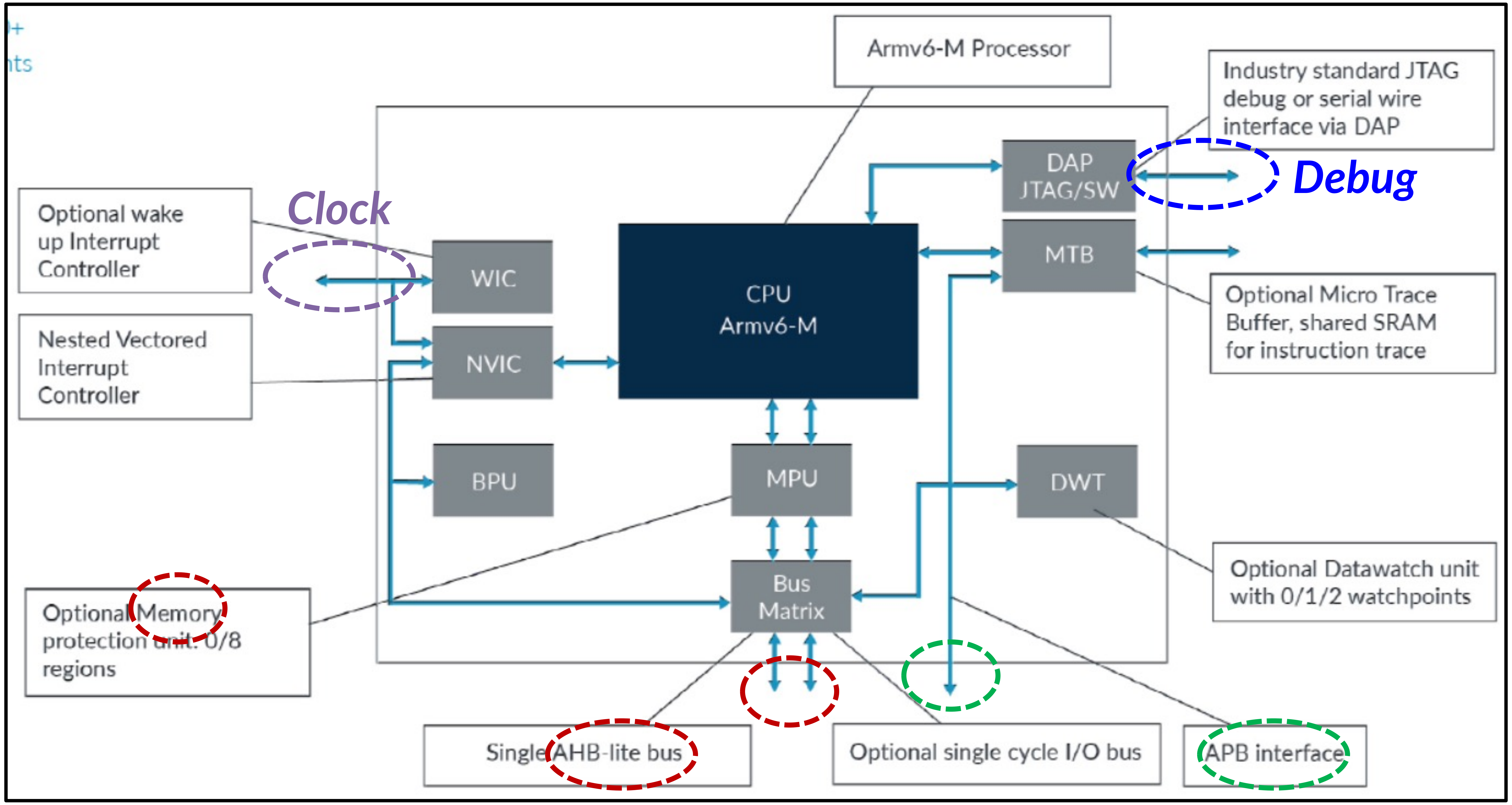
Block Diagram



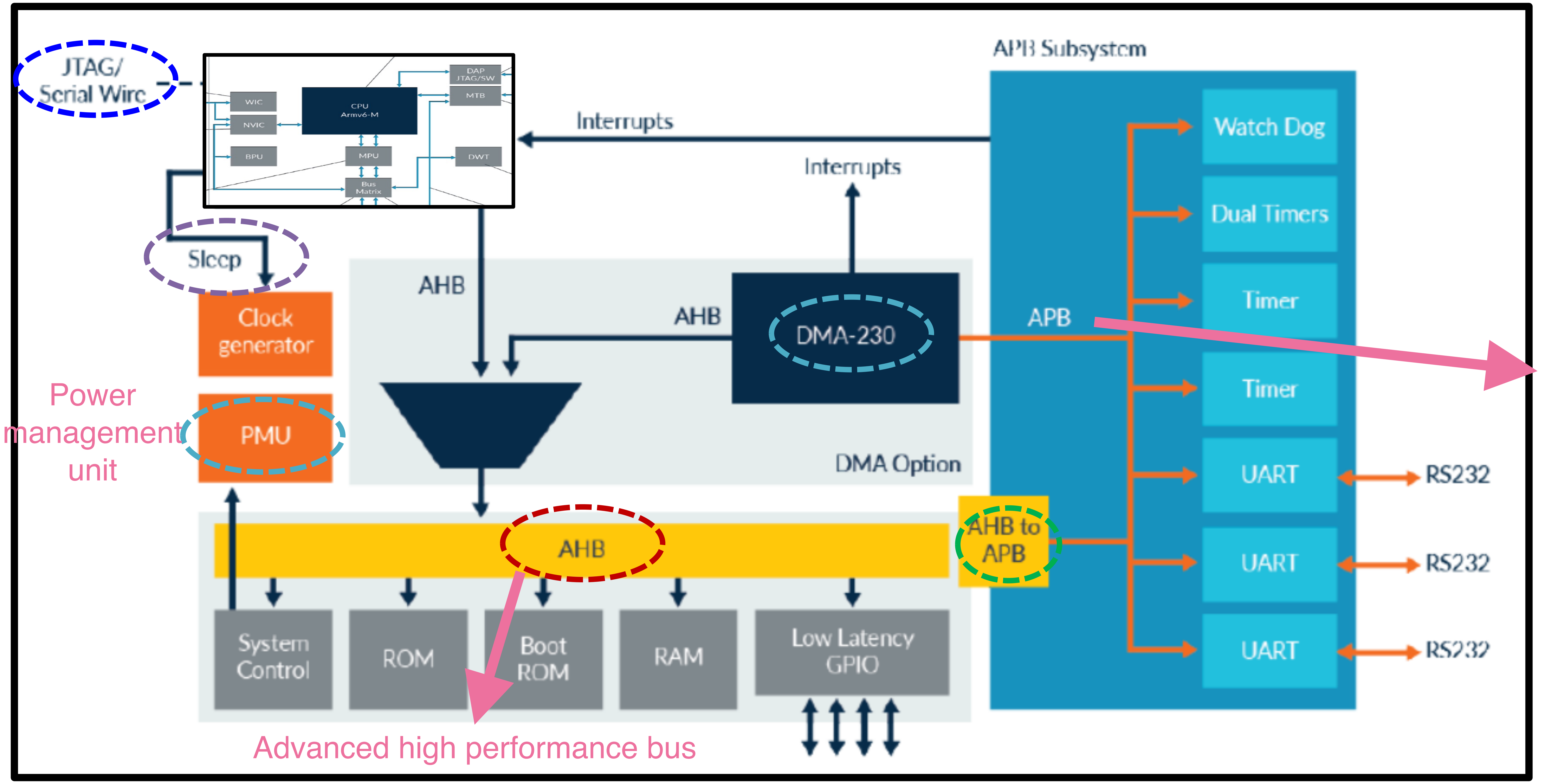
**Advanced High-performance Bus (AHB)
**Advanced Peripheral Bus (APB)

ARM Cortex M0+

Block Diagram

Any Familiar Components??

Does this diagram look familiar???



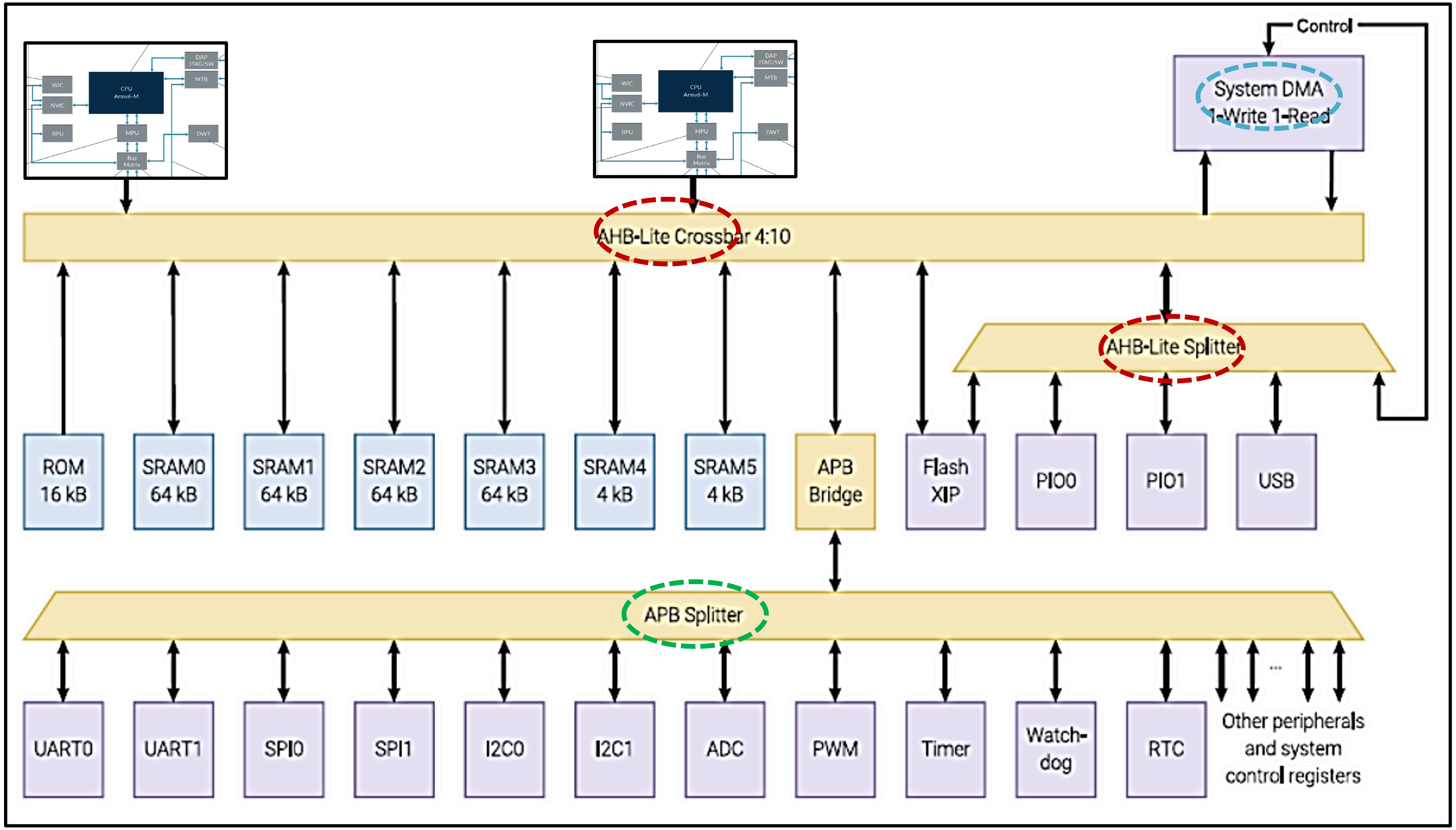
Advanced Peripheral Bus

Advanced high performance bus

designed to facilitate high-speed communication between various components in a microcontroller or system-on-chip (SoC).

**Advanced High-performance Bus (AHB)
**Advanced Peripheral Bus (APB)

Block Diagram



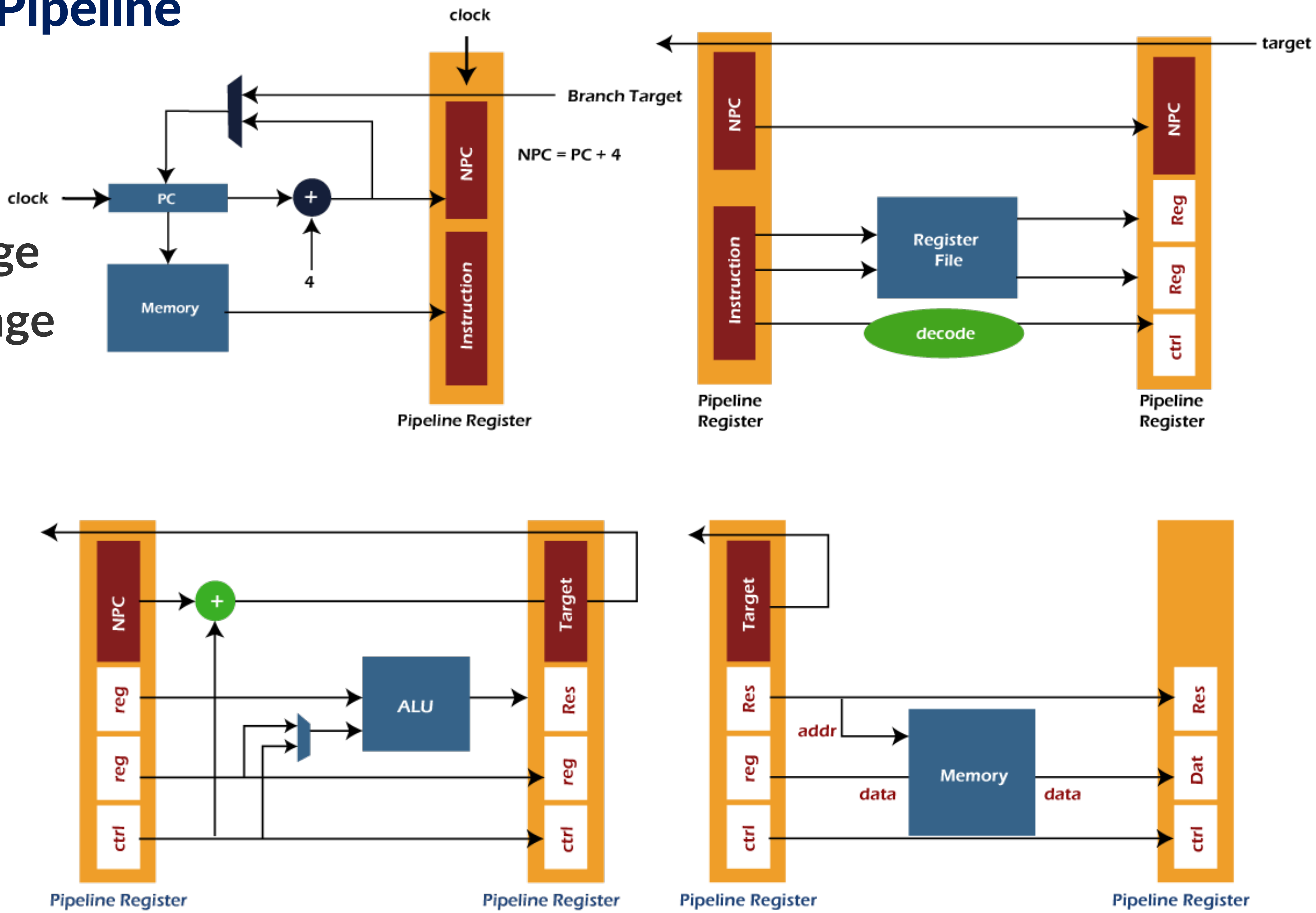
manages communication between the APB bus and multiple peripheral devices.

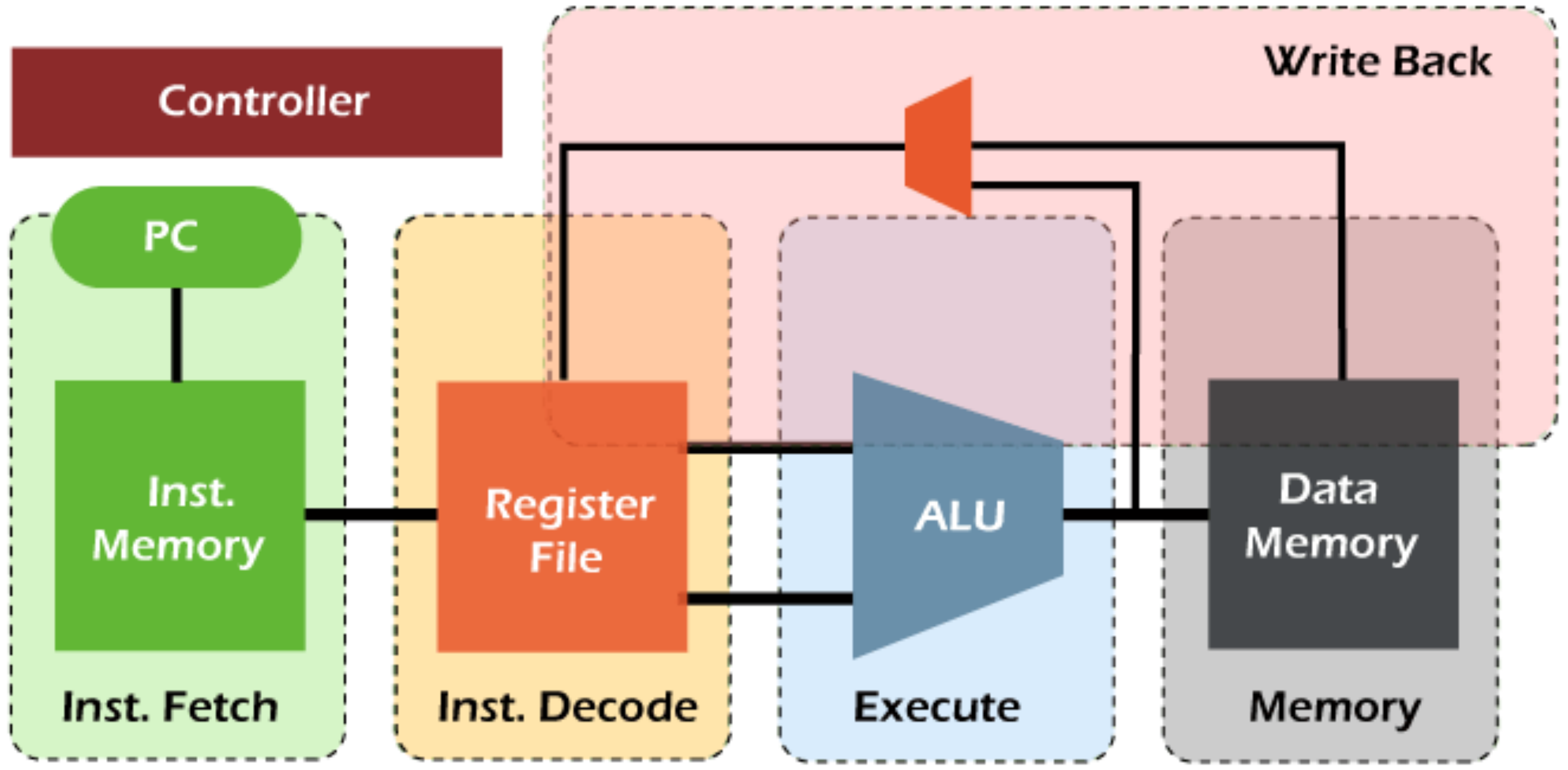
Execution, Stages and Throughput in Pipeline

- Pipelining can be described as a technique in which **multiple instructions are overlapped at the time of execution.**
- There are **five stages** in the pipeline, and all these stages are connected with each other so that they can form a pipe-like structure.
- In the pipeline, there are **two ends** in which one end is used to **enter** the instruction, and the second end is used to **exit** it.
- Because of the pipeline, the overall **instruction throughput** is **increased**.
- Each segment of the pipeline system has the input register followed by a combinational circuit.
- The data is contained with the help of a register, and operations on that data are performed with the help of this combinational circuit.
- The result of a combinational circuit will be applied to the input register of next segment.

Execution, Stages and Throughput in Pipeline

- The five stages are:
  - ➢Stage 1 is the **instruction fetch**
  - ➢Stage 2 is the **instruction decodes stage**
  - ➢Stage 3 is the **Instruction executes stage**
  - ➢Stage 4 is the **Memory Access stage**
  - ➢Stage 5 is the **Write Back stage**





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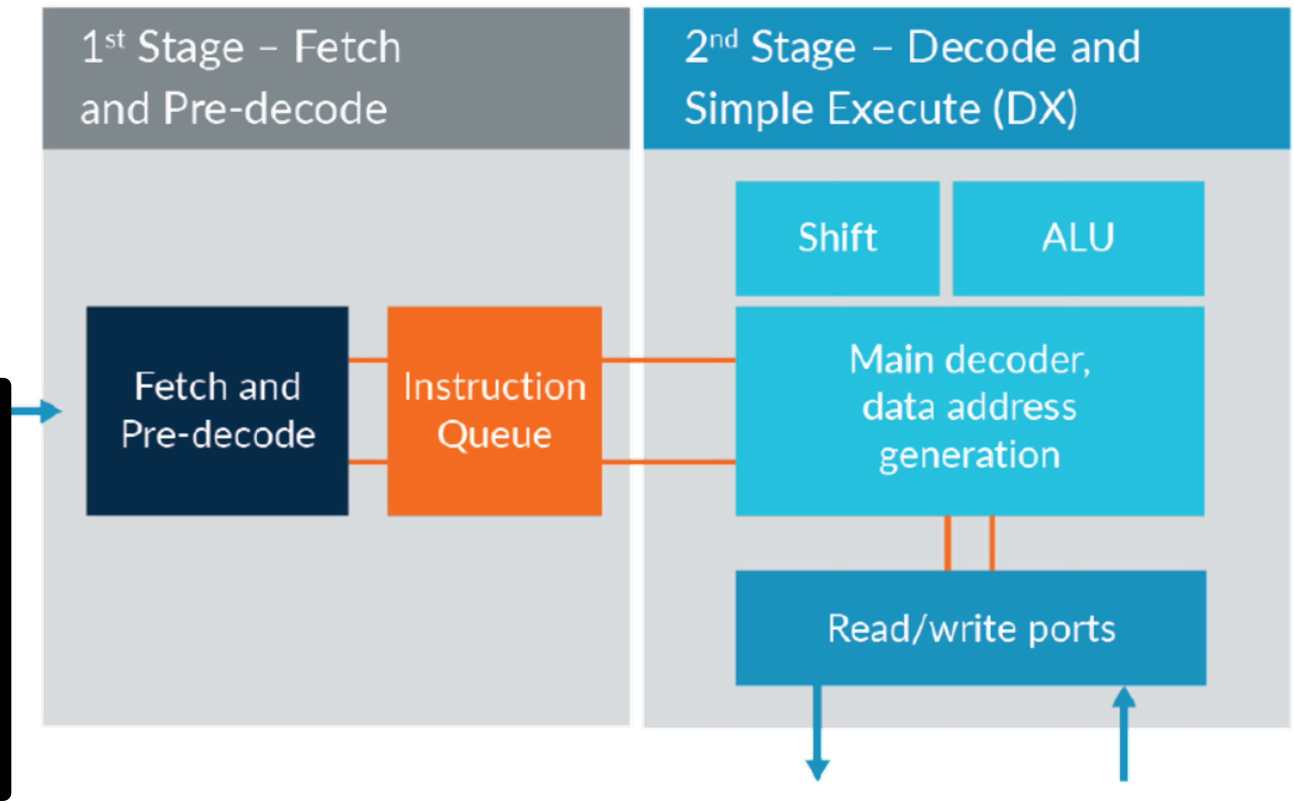
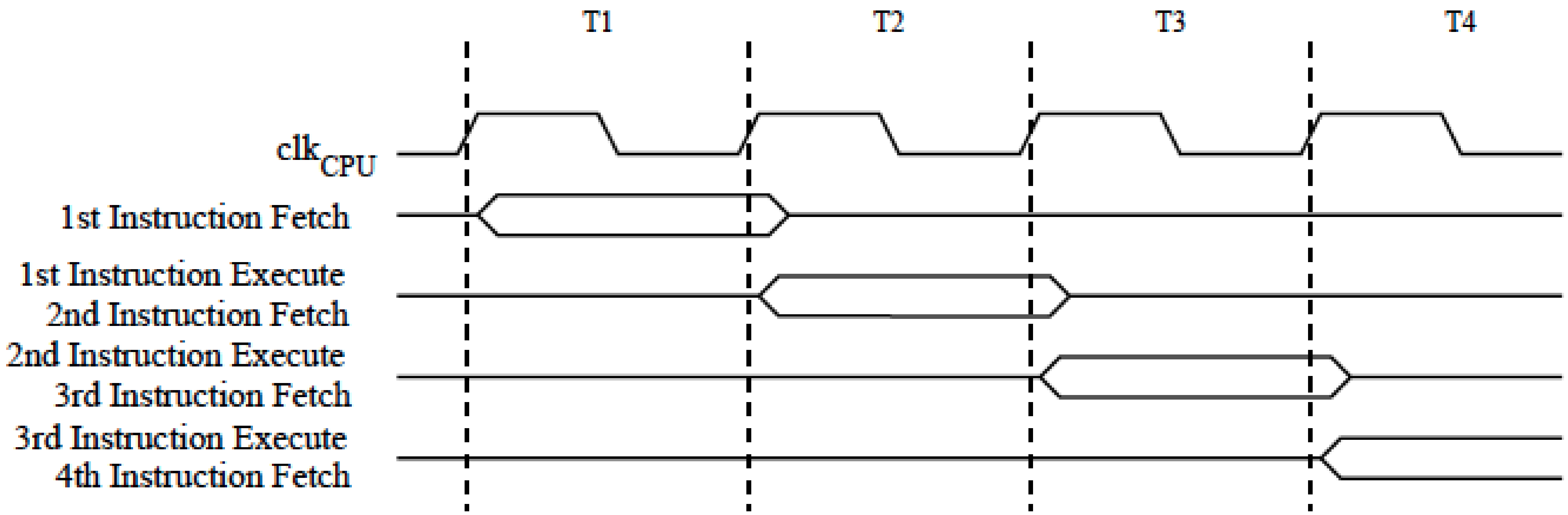
Execution Pipelining Types

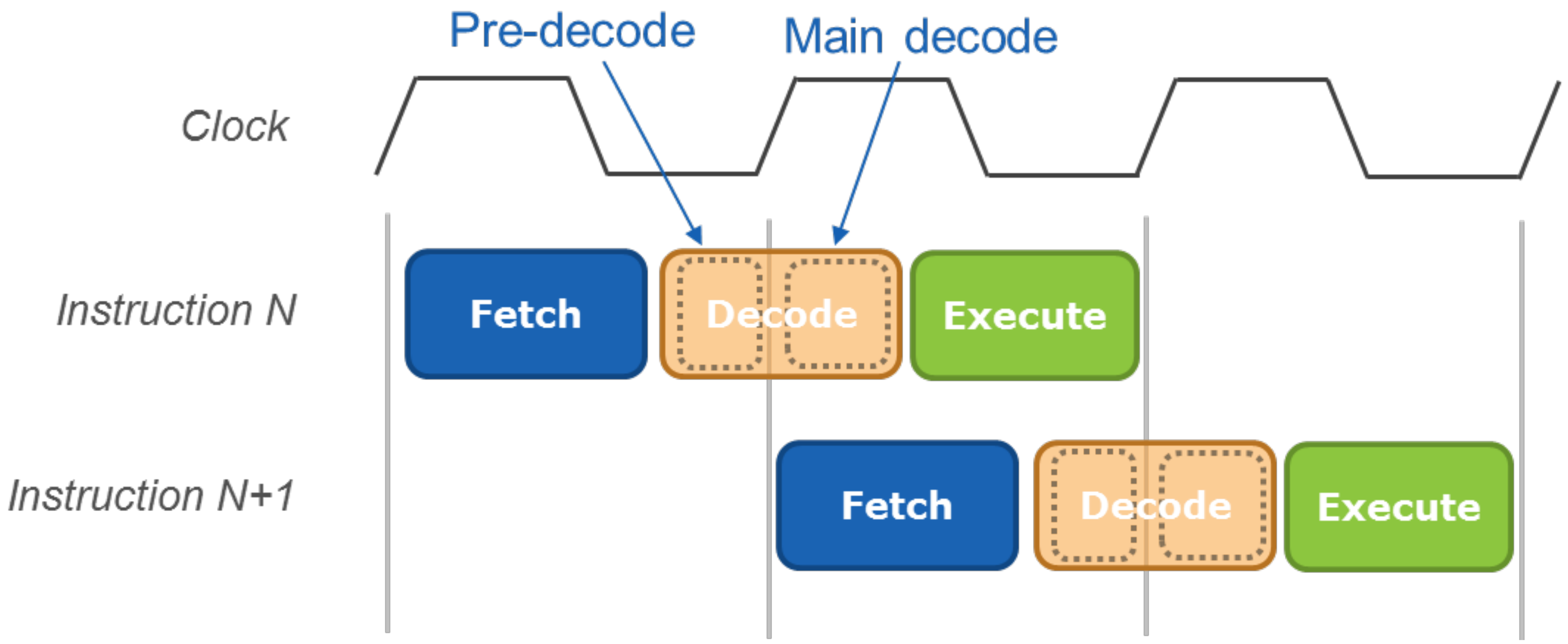
The more pipe stages there are, the faster the pipeline



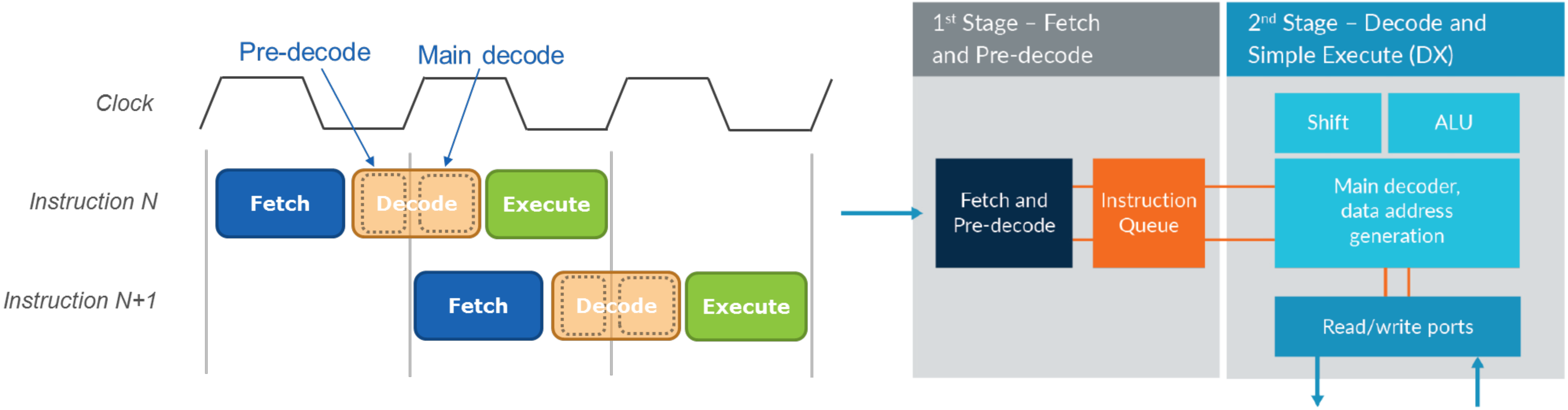
Single Level Pipelining



2-Level Pipelining

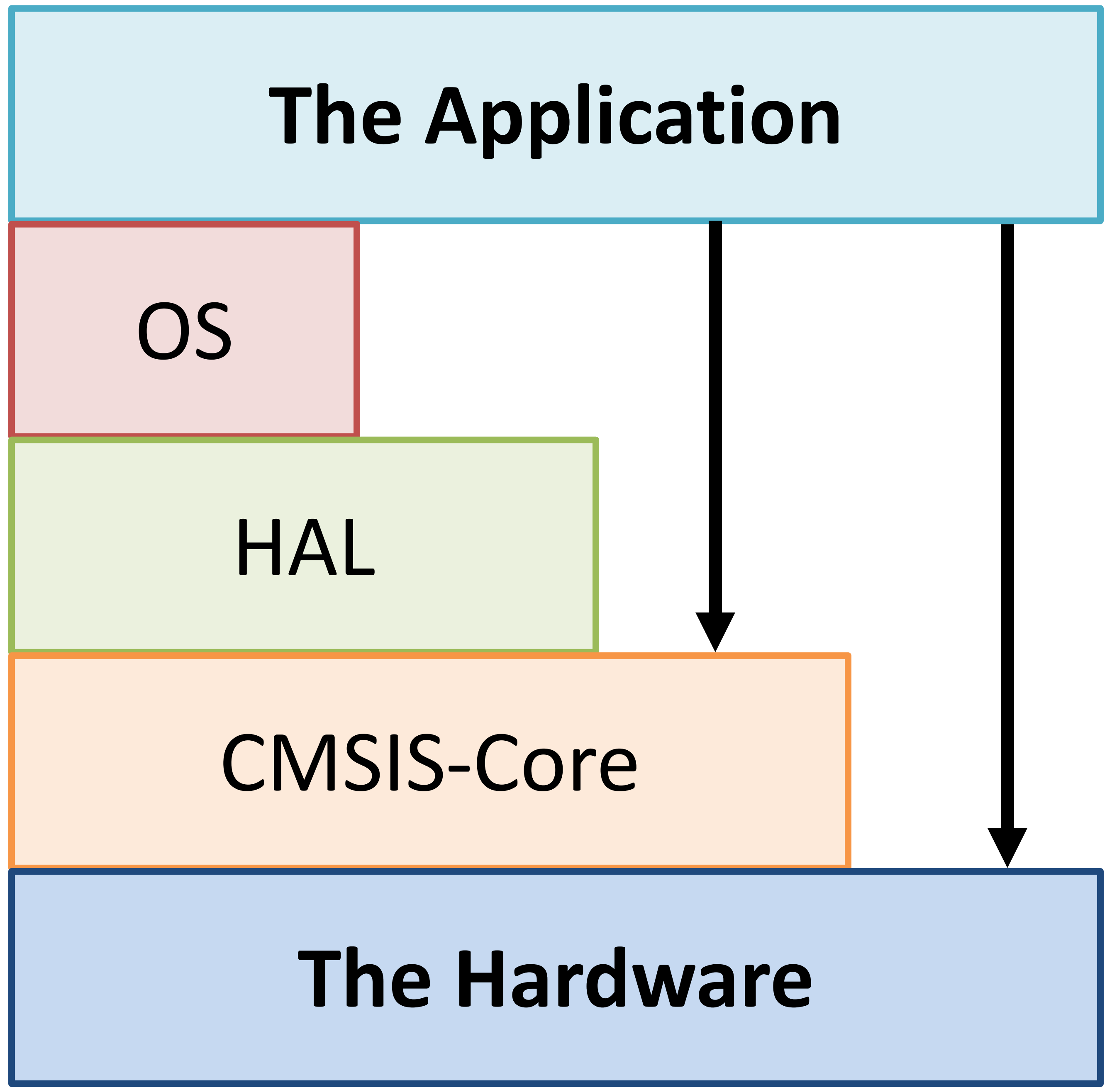


Pipelining


Pre-decode
Main decode
Clock
Instruction N
Fetch
Decode
Execute
Instruction N+1
Fetch
Decode
Execute
1st Stage – Fetch and Pre-decode
2nd Stage – Decode and Simple Execute (DX)
Shift
ALU
Fetch and Pre-decode
Instruction Queue
Main decoder, data address generation
Read/write ports

Programming Tools

- Visual wizard tools (such as STMCubeMX, Arduino IDE) *→ Let's see this together now ☺*
- Hardware Abstraction Layer (HAL) libraries *→ For the rest of the semester ☺*
- Bare-metal: Bypass HAL and possibly CMSIS-Core *→ Next Lecture & Tutorials ☺*



**Common Microcontroller Software Interface Standard (CMSIS)

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Let's have some fun ☺

For Further Inquiries, Please send an email

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Thank you for your attention!

See you next time ☺